A Major Project Report
(Phase-I)

On

AI-Based Remote Patient Monitoring and Early Disease Prediction System

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By

Dev Ranjan Raut (2410031623)

Ritesh Rajpurohit (2410031626)

Simmi Tiwari (2410031627)

Mehul Arora (2410031628)

Under the guidance of

Dr. Rani Kumari (Assistant Professor)

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
IILM UNIVERSITY, GREATER NOIDA
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SCHOOL OF COMPUTER SCIENCE AND ENGG.

IILM UNIVERSITY, GREATER NOIDA

CANDIDATE'S DECLARATION

We hereby certify that the work presented in the Phase-I report of the project entitled "AI-Based Remote Patient Monitoring and Early Disease Prediction System", in partial fulfillment of the requirements for the degree of Bachelor of Technology in the School of Computing Science and Engineering, IILM University, Greater Noida, is an original work carried out by us under the supervision of Dr. Rani Kumari, School of Computing Science and Engineering, IILM University, Greater Noida.

The work presented in this Phase-I report has not been submitted by us for the award of any other degree or qualification at this or any other institution.

- Dev Ranjan Raut (2410031623)
- Ritesh Rajpurohit (2410031626)
- Simmi Tiwari (2410031627)
- Mehul Arora (2410031628)

This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

Signature of Guide Name: Dr. Rani Kumari

Designation: Assistant Professor

School: School of Computing Science and Engineering,

IILM University, Greater Noida

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1. Abstract

The convergence of artificial intelligence and digital healthcare presents a transformative opportunity to shift medical paradigms from reactive treatments to proactive interventions. The "AI-Based Remote Patient Monitoring and Early Disease Prediction System" is a comprehensive, web-based healthcare platform designed to serve as an intelligent, early-warning decision-support tool. Traditional remote monitoring systems and diagnostic tools predominantly rely on static, point-in-time clinical data, failing to account for the longitudinal deterioration or improvement in a patient's health trajectory. This project addresses this critical gap by integrating static machine learning predictive models—targeting Cardiovascular Disease, Diabetes, Chronic Kidney Disease (CKD), and Liver Disease—with a dynamic longitudinal monitoring simulator.

The system actively evaluates patient profiles and continuous health measurements to establish personalized health baselines. By calculating trend deviations from these baselines, the Alert Engine generates smart, prioritized warnings before critical physiological thresholds are breached. Furthermore, to bridge the trust gap often associated with "black-box" AI models, the system incorporates Shapley Additive exPlanations (SHAP). This Explainable AI (XAI) layer translates complex algorithmic outputs into intuitive, human-readable insights for both patients and healthcare providers. Delivered through tailored React-based dashboards and powered by a robust FastAPI backend, this software-centric solution aims to enhance clinical decision-making, reduce alert fatigue, and empower patients with actionable health intelligence.

2. Problem Definition

Despite the rapid digitalization of healthcare records, current patient monitoring and disease prediction systems suffer from several structural and algorithmic limitations that hinder effective early intervention:

Over-reliance on Static Data: Most existing clinical risk prediction models evaluate a patient's health based on isolated, periodic check-ups (one row of data per patient). They fail to capture the *velocity* of change (e.g., a blood pressure reading that is technically within normal limits but is rapidly increasing over a 7-day period).

High False-Alarm Rates and Alert Fatigue: Traditional Remote Patient Monitoring (RPM) systems rely on rigid, hardcoded clinical thresholds (e.g., generating an alert every time Heart Rate > 100 bpm). This lack of personalization leads to excessive alerts, causing physicians to experience "alert fatigue" and potentially ignore critical warnings.

The "Black-Box" Nature of AI in Healthcare: Advanced machine learning models (like XGBoost or Neural Networks) offer high accuracy but operate as black boxes.

When a system predicts a 75% risk of Diabetes, it traditionally provides no clinical justification. Without transparency, medical professionals cannot trust the AI's recommendation, and patients cannot take targeted corrective actions.

Fragmented Ecosystems: Predictive modeling and continuous remote monitoring are often treated as separate research domains. There is a lack of unified platforms that seamlessly combine disease-specific risk assessment with continuous vital-sign tracking.

3. Literature Survey

A comprehensive review of recent literature spanning 2021 to 2026 highlights the evolution of remote healthcare and the prevailing gaps in predictive analytics:

Remote Patient Monitoring and IoT (Smith et al., 2023): Recent studies emphasize the proliferation of wearable health technology. Smith et al. demonstrated that continuous monitoring of vitals (SpO2, HR, BP) significantly reduces hospital readmission rates. However, the study noted that raw data streams overwhelm clinical staff without an intelligent filtering layer [1].

Ensemble Learning for Chronic Disease Prediction (Patel & Sharma, 2024): Patel and Sharma evaluated various machine learning algorithms on standardized clinical datasets (e.g., UCI Machine Learning Repository). Their findings established that ensemble methods, specifically XGBoost and Random Forest, outperform traditional logistic regression in handling tabular clinical data with non-linear relationships and missing values (common in CKD and Liver datasets) [2].

The Need for Longitudinal AI Models (Chen et al., 2025): Chen et al. identified a critical research gap: most AI models are trained on cross-sectional data. Their research advocated for integrating time-series analysis (trend lines and slopes) with static risk probabilities to create dynamic risk trajectories, a concept foundational to our proposed system's Alert Engine [3].

Explainable AI in Clinical Settings (Wang et al., 2025): The integration of SHAP (Shapley Additive exPlanations) in healthcare has been widely researched to decode complex models. Wang et al. concluded that providing feature-importance graphs directly to clinicians increased AI adoption rates by 40%, as it aligns the algorithm's reasoning with established clinical pathways [4].

Conclusion of Survey: While individual technologies (RPM, Ensemble ML, XAI) are well-researched, there is a distinct lack of open-source, integrated software architectures that combine these elements into a unified, patient-and-doctor-facing decision support system. This project directly addresses this synthesis gap.

4. Objectives

The primary objective of this project is to develop a holistic, software-based early-warning healthcare platform. The specific sub-objectives are:

Data Aggregation & Management: To develop a secure, relational database architecture (using SQLite/PostgreSQL) capable of storing patient demographic profiles, medical histories, and high-frequency health measurements.

Predictive Modeling: To train, optimize, and deploy four separate machine learning models (preferably XGBoost) for evaluating the risk of Cardiovascular Disease, Diabetes, Chronic Kidney Disease, and Liver Disease using established Kaggle/UCI clinical datasets.

Longitudinal Trend Analysis: To engineer a Patient Monitoring Simulator and Trend Engine that calculates personalized physiological baselines and computes deviation metrics (slopes and moving averages) over time.

Smart Alert Generation: To design a heuristic Alert Engine that mathematically combines the static ML risk score with longitudinal trend deviations to output prioritized alert levels (Green, Amber, Red).

Explainability (XAI): To integrate SHAP values to dynamically generate natural language explanations for patients and detailed feature-impact visualizations for doctors.

User Interface Design: To build distinct, intuitive React-based dashboards for both patients (focusing on readability and trends) and medical professionals (focusing on triage and deep clinical analytics).

5. Scope

To ensure a focused and feasible product development lifecycle, the project boundaries are strictly defined:

In-Scope:

- Development of a full-stack web application (React frontend, FastAPI backend).
- Implementation of 4 supervised machine learning models for specific chronic diseases.
- Development of a simulation layer to ingest synthetic longitudinal wearable data (to mock continuous monitoring).
- Implementation of SHAP for model explainability.
- Deployment of role-based access control (Patient vs. Doctor dashboards).

Out-of-Scope (Future Enhancements):

- Hardware development (e.g., building custom IoT sensors, ESP32 boards, or smartwatch hardware).
- Live integration with physical medical devices or proprietary APIs (e.g., Apple

HealthKit, Fitbit API).
Deep Learning architectures (CNNs/LSTMs) for raw ECG/EEG signal processing.
Federated Learning or Edge AI deployments.
Official medical device certification (FDA/CE) or autonomous clinical diagnosis.

6. Novelty

The innovation of this project lies in its architectural design and data-fusion approach rather than algorithmic reinvention:

Dynamic Risk Trajectory: Instead of providing a static "65% Risk" output, the system provides a trajectory (e.g., "Stable → Stable → Increasing → High"). It fuses static ML model outputs with time-series mathematical trends to catch deterioration early.

Contextualized Alerts: The system reduces alert fatigue by recognizing personalized baselines. A heart rate of 95 bpm might trigger a generic system's alarm, but this system will evaluate if 95 bpm is the patient's normal baseline before escalating the alert.

Dual-Layer Explainability: It provides bifurcated XAI—translating SHAP values into simple textual insights for the patient ("Your risk is elevated mainly due to increasing glucose"), while providing standard, granular SHAP dependency plots for the doctor.

7. Feasibility Study

A rigorous feasibility analysis confirms that the proposed system is practical, viable, and achievable within the academic timeline.

A. Technical Feasibility

The chosen technology stack (React, Python, FastAPI, XGBoost, SHAP, SQLite) is highly robust, heavily documented, and supported by massive open-source communities. The models will be trained on standard, pre-cleaned academic datasets (UCI/Kaggle), which eliminates the bottleneck of primary medical data collection. Local development environments (such as Apple Silicon MacBooks) possess more than enough computational power to train these tabular models and run the FastApi server concurrently without latency issues.

B. Operational Feasibility

Operationally, the system is designed to seamlessly integrate into a conceptual clinical workflow. The dual-dashboard design ensures that patients are not overwhelmed with complex medical jargon, while doctors are provided with efficient triage tools (sorting patients by highest risk and deteriorating trends). The RESTful API architecture (via FastAPI) ensures that the backend can easily be connected to a mobile application in future phases.

C. Multidisciplinary & Transdisciplinary Aspects

This project successfully bridges multiple academic disciplines:
Software Engineering: Structuring a decoupled client-server architecture, database normalization, and secure API routing.
Data Science & AI: Data wrangling, feature engineering, hyperparameter tuning, model evaluation (ROC-AUC, F1-scores), and Explainable AI implementation.
Health Informatics: Understanding clinical parameters, modeling disease risk factors based on medical literature, and designing ethical, decision-support workflows that prioritize patient safety.

D. Identification of Components and Bill of Materials (BoM)

Adhering to the strictly software-based scope, all required tools, libraries, and datasets are open-source. This yields a highly optimized product development scope with a total material cost of zero.

S.No.	Component / Technology	Purpose	License / Cost
1	React.js, Vite, Tailwind CSS	Frontend UI/UX development and Dashboard rendering.	Open Source (₹0)
2	Python 3, FastAPI, Uvicorn	Backend API server and business logic routing.	Open Source (₹0)
3	Scikit-learn, XGBoost, Pandas	Data preprocessing and Machine Learning	Open Source (₹0)

S.No.	Component / Technology	Purpose	License / Cost
		model training.	
4	SHAP (SHapley Additive exPlanations)	Generating feature importance and model explainability.	Open Source (₹0)
5	SQLite (Phase 1) / PostgreSQL	Relational database for storing users, patients, and logs.	Open Source (₹0)
6	Kaggle/UCI Clinical Datasets	Training data for CVD, Diabetes, CKD, and Liver models.	Public Domain (₹0)
7	Synthetic Wearable Dataset	Longitudinal data for the Patient Monitoring Simulator.	Public Domain (₹0)

Total Estimated Hardware/Software Cost: ₹0.00 (Development executed on existing personal student workstations).

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